Chapter 5 Breakout (Pt. III)

STAT 4714

Questions

1. The probability that a person living in a certain city owns a dog is estimated to be 0.3. Find the probability that the tenth person randomly interviewed in that city is the fifth one to own a dog.

$p = 0.3$ $\quad$ $X = 10$ $\quad$ $k = 5$

# of trials required to find k successes ≡ negative binomial

$$nb(x; k=5, p=0.3)$$

$$P(X=10) = \binom{10-1}{5-1}(0.3)^5(0.7)^5$$

or

$$= \text{dnbinom}(10-5, 5, 0.3)$$

$$= 0.0515$$

2. Three people toss a fair coin and the odd one pays for coffee. If the coins all turn up the same, they are tossed again. Find the probability that fewer than 4 tosses are needed.

interested in # of trials needed for first "success," where "success" is that one person flipped differently than the other two.

$\Rightarrow$ geometric experiment

$p(\text{all flips are same}) = 2 \cdot (1/2)^3 = 1/4$, so $\underbrace{p = 1 - 1/4 = 3/4}_{\text{probability of "success"}}$

$g(x; p = 3/4)$

$$P(X<4) = P(X \le 3) = \sum_{x=1}^{3} (0.75)(1-0.25)^{x-1}$$

or

$= \text{pgeom}(3-1, 0.75)$

$= 0.9844$

3. The probability that a student pilot passes the written test for a private pilot's license is 0.7. Find the probability that a given student will pass the test

$p = 0.7$

(a) on the third try;

(b) before the fourth try.

we're looking for the first success $\Rightarrow$ geometric

$$g(x; 0.7)$$

X = # of trials until first success

(a) $P(X=3) = g(3; 0.7) = 0.7(0.3)^{3-1}$

or

$= \text{dgeom}(3-1, 0.7) = \underline{0.063}$

(b) $P(X<4) = P(X \leq 3)$

$$= \sum_{x=1}^{3} 0.7(0.3)^{x-1}$$

or

$= \text{pgeom}(3-1, 0.7) = \underline{0.973}$

Poisson!

4. On average, a textbook author makes two word-processing errors per page on the first draft of their textbook. What is the probability that on the next page they will make

   (a) 4 or more errors?

   (b) no errors?

$\lambda = 2$

$X = \#$ of errors

$$p(x; \lambda = 2) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots$$

(a) $P(X \geq 4) = 1 - P(X \leq 3)$

$$= 1 - \sum_{x=0}^{3} \frac{e^{-\lambda}\lambda^x}{x!}$$

or

$$= 1 - \text{ppois}(3, \lambda = 2) = \underline{0.1429}$$

(b) $P(X = 0) = \dfrac{e^{-2}\, 2^0}{0!} = e^{-2} = \underline{0.1353}$

Poisson!

5. The average number of field mice per acre in a 5-acre wheat field is estimated to be 12. Find the probability that fewer than 7 field mice are found

$\lambda = 12$

(a) on a given acre;

(b) on 2 of the next 3 acres inspected.

$X = \#$ of mice found per acre

$$p(x; \lambda = 12) = \frac{e^{-12} 12^x}{x!}, \quad x = 0, 1, 2, \ldots$$

(a) $P(X < 7) = P(X \le 6) = \sum_{x=0}^{6} \frac{e^{-12} 12^x}{x!}$

or

$= \text{ppois}(6, 12) = \underline{0.0458}$

(b) prop for each acre is $0.0458 = p$

inspect 3 acres $\Rightarrow n = 3$

$\Rightarrow$ Binomial experiment

$b(x; n = 3, p = 0.0458)$

$P(X = 2) = \text{dbinom}(2, 3, 0.0458)$

$= \underline{0.0060}$

6. The probability that a person will die when they contract a virus infection is 0.001. Of the next 4000 people infected, what is the mean number who will die?

$p = 0.001$

$n = 4000$

X = # of people who die

Binomial experiment!

$\Rightarrow E(X) = np = 4000(0.001) = 4$